

MH1201 Linear Algebra II

Tutorial 1 (Week 2) – Problems & Solutions

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Overview of This Tutorial Problem Set

This sheet focuses on core *subspace* ideas and common counterexamples:

- Subspaces defined by linear constraints in $\mathbb{R}^n$.
- Function spaces and subspaces defined by linear conditions (evaluation/derivative).
- Additive subgroups that fail scalar closure.
- Unions of subspaces: when they are (and are not) subspaces.
- Subsets of matrix spaces and quick subspace tests.
- Direct sums and constructing complementary subspaces in $\mathbb{R}^n$.

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Question 1 (Linear constraint in $\mathbb{R}^3$)

Problem

Determine whether

$$\{(a, b, c) \in \mathbb{R}^3 \mid a = 5c\}$$

is a vector subspace of $\mathbb{R}^3$.

Solution

Method 1: Subspace test (axioms)

Let

$$U := \{(a, b, c) \in \mathbb{R}^3 \mid a = 5c\}.$$

(i) Zero vector. The zero vector $(0, 0, 0)$ satisfies $0 = 5 \cdot 0$, hence $(0, 0, 0) \in U$.

(ii) Closure under addition. Take $(a_1, b_1, c_1), (a_2, b_2, c_2) \in U$. Then $a_1 = 5c_1$ and $a_2 = 5c_2$. Therefore

$$a_1 + a_2 = 5c_1 + 5c_2 = 5(c_1 + c_2),$$

so $(a_1 + a_2, b_1 + b_2, c_1 + c_2) \in U$.

(iii) Closure under scalar multiplication. Take $(a, b, c) \in U$ and $\lambda \in \mathbb{R}$. Since $a = 5c$,

$$\lambda a = \lambda(5c) = 5(\lambda c),$$

so $(\lambda a, \lambda b, \lambda c) \in U$.

Thus U is a subspace of $\mathbb{R}^3$, and the final answer is

$U \text{ is a vector subspace of } \mathbb{R}^3.$

Method 2: Parametrisation / spanning set

The condition $a = 5c$ means that any vector in U has the form

$$(a, b, c) = (5c, b, c) = b(0, 1, 0) + c(5, 0, 1).$$

Hence

$$U = \text{span}\{(0, 1, 0), (5, 0, 1)\}.$$

A span of vectors is always a subspace, so again

$U \text{ is a vector subspace of } \mathbb{R}^3.$

Question 2 (Subspace of differentiable functions)

Problem

Recall that $\mathbb{R}^{\mathbb{R}}$ is a vector space under pointwise addition and pointwise scalar multiplication. Show that the set of differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f'(1) = 3f(2)$$

is a vector subspace of $\mathbb{R}^{\mathbb{R}}$.

Solution

Let $\mathcal{D}$ be the set of all differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$, and define

$$W := \{f \in \mathcal{D} \mid f'(1) = 3f(2)\}.$$

Since $\mathcal{D} \subseteq \mathbb{R}^{\mathbb{R}}$ is itself a vector space under pointwise operations, it suffices to show W is a subspace of $\mathcal{D}$.

Method 1: Direct subspace verification

(i) Zero function. Let 0 denote the zero function $0(x) = 0$. Then $0'(1) = 0$ and $3 \cdot 0(2) = 0$, so $0 \in W$.

(ii) Closure under addition. Take $f, g \in W$. Then $f'(1) = 3f(2)$ and $g'(1) = 3g(2)$. Because $(f + g)' = f' + g'$, we get

$$(f + g)'(1) = f'(1) + g'(1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3(f + g)(2).$$

So $f + g \in W$.

(iii) Closure under scalar multiplication. Take $f \in W$ and $\lambda \in \mathbb{R}$. Because $(\lambda f)' = \lambda f'$, we get

$$(\lambda f)'(1) = \lambda f'(1) = \lambda \cdot 3f(2) = 3(\lambda f(2)) = 3(\lambda f)(2).$$

So $\lambda f \in W$.

Therefore W is a vector subspace, and the final answer is

$W \text{ is a vector subspace of } \mathbb{R}^{\mathbb{R}}.$

Method 2: Kernel of a linear functional

Define a map $L : \mathcal{D} \rightarrow \mathbb{R}$ by

$$L(f) := f'(1) - 3f(2).$$

We check linearity. For $f, g \in \mathcal{D}$ and $\lambda \in \mathbb{R}$,

$$\begin{aligned} L(f + g) &= (f + g)'(1) - 3(f + g)(2) \\ &= (f'(1) + g'(1)) - 3(f(2) + g(2)) \\ &= (f'(1) - 3f(2)) + (g'(1) - 3g(2)) \\ &= L(f) + L(g), \end{aligned}$$

and

$$\begin{aligned} L(\lambda f) &= (\lambda f)'(1) - 3(\lambda f)(2) \\ &= \lambda f'(1) - 3\lambda f(2) \\ &= \lambda(f'(1) - 3f(2)) \\ &= \lambda L(f). \end{aligned}$$

Hence L is linear. Now observe:

$$W = \{f \in \mathcal{D} \mid f'(1) = 3f(2)\} = \{f \in \mathcal{D} \mid L(f) = 0\} = \ker(L).$$

The kernel of a linear map is a subspace, so

W is a vector subspace of $\mathbb{R}^{\mathbb{R}}$.

Question 3 (Additive closure but not scalar closure)

Problem

Find a subset $S \subseteq \mathbb{R}^2$ that satisfies the following properties:

- S is closed under vector addition and under taking additive inverses.
- S is not closed under scalar multiplication.

Solution

Method 1: Integer lattice example

Let

$$S := \mathbb{Z}^2 = \{(m, n) \in \mathbb{R}^2 \mid m, n \in \mathbb{Z}\}.$$

Closed under addition. If $(m_1, n_1), (m_2, n_2) \in \mathbb{Z}^2$, then

$$(m_1, n_1) + (m_2, n_2) = (m_1 + m_2, n_1 + n_2) \in \mathbb{Z}^2$$

since sums of integers are integers.

Closed under additive inverses. If $(m, n) \in \mathbb{Z}^2$, then

$$-(m, n) = (-m, -n) \in \mathbb{Z}^2$$

since negatives of integers are integers.

Not closed under scalar multiplication. Take $(1, 0) \in \mathbb{Z}^2$ and scalar $\lambda = \frac{1}{2} \in \mathbb{R}$. Then

$$\frac{1}{2}(1, 0) = \left(\frac{1}{2}, 0\right) \notin \mathbb{Z}^2.$$

Hence $\mathbb{Z}^2$ fails scalar closure (over $\mathbb{R}$), so it is not a vector subspace.

Thus one valid answer is

$$\boxed{S = \mathbb{Z}^2}.$$

Method 2: Rational lattice example

Alternatively, let

$$S := \mathbb{Q}^2 = \{(p, q) \in \mathbb{R}^2 \mid p, q \in \mathbb{Q}\}.$$

Closed under addition and additive inverses. If $p_1, p_2, q_1, q_2 \in \mathbb{Q}$, then $p_1 + p_2 \in \mathbb{Q}$, $q_1 + q_2 \in \mathbb{Q}$, and $-p_1, -q_1 \in \mathbb{Q}$. So $\mathbb{Q}^2$ is closed under addition and taking negatives.

Not closed under scalar multiplication by reals. Take $(1, 0) \in \mathbb{Q}^2$ and scalar $\lambda = \sqrt{2} \in \mathbb{R}$. Then

$$\sqrt{2}(1, 0) = (\sqrt{2}, 0) \notin \mathbb{Q}^2.$$

So another valid answer is

$$\boxed{S = \mathbb{Q}^2}.$$

Question 4 (Union of subspaces may fail)

Problem

Find vector subspaces W_1 and W_2 of $\mathbb{R}^2$ such that $W_1 \cup W_2$ is not a vector subspace of $\mathbb{R}^2$.

Hint: Consider the coordinate axes of $\mathbb{R}^2$.

Solution

Method 1: Coordinate axes counterexample

Let

$$W_1 = \{(x, 0) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}, \quad W_2 = \{(0, y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\}.$$

Each is a subspace (each is a line through the origin).

Now $(1, 0) \in W_1 \subseteq W_1 \cup W_2$ and $(0, 1) \in W_2 \subseteq W_1 \cup W_2$. But

$$(1, 0) + (0, 1) = (1, 1) \notin W_1 \cup W_2$$

since $(1, 1)$ lies on neither axis. So $W_1 \cup W_2$ is not closed under addition and hence is not a subspace.

Therefore one valid choice is

$W_1 = \{(x, 0)\}, \quad W_2 = \{(0, y)\} \text{ and } W_1 \cup W_2 \text{ is not a subspace.}$

Method 2: Conceptual reason (two distinct lines)

Let W_1, W_2 be two distinct 1-dimensional subspaces (lines through the origin) in $\mathbb{R}^2$, with $W_1 \neq W_2$. Choose $u \in W_1 \setminus W_2$ and $v \in W_2 \setminus W_1$; such vectors exist because the subspaces are distinct. Then u, v are linearly independent (if u were a scalar multiple of v , it would lie in W_2). Hence $u+v$ is not in W_1 (otherwise $v = (u+v) - u \in W_1$), and similarly $u+v \notin W_2$. Thus $u, v \in W_1 \cup W_2$ but $u+v \notin W_1 \cup W_2$, so the union cannot be a subspace.

Applying this to the x -axis and y -axis reproduces Method 1.

Question 5 (When is $W_1 \cup W_2$ a subspace?)

Problem

Let V be a vector space, and let W_1 and W_2 be vector subspaces of V . Show that $W_1 \cup W_2$ is a vector subspace of V if and only if either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Hint: Use the Principle of Contradiction to establish the forward direction.

Solution

Method 1: Direct contradiction (subspace-axiom argument)

($\Rightarrow$) Assume $W_1 \cup W_2$ is a subspace of V . Suppose for contradiction that neither inclusion holds. Then there exist

$$u \in W_1 \setminus W_2, \quad v \in W_2 \setminus W_1.$$

Since $u, v \in W_1 \cup W_2$ and $W_1 \cup W_2$ is a subspace, it is closed under addition, hence

$$u + v \in W_1 \cup W_2.$$

If $u + v \in W_1$, then $v = (u + v) - u \in W_1$, contradicting $v \notin W_1$. If $u + v \in W_2$, then $u = (u + v) - v \in W_2$, contradicting $u \notin W_2$. Therefore the assumption was false, and thus $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

($\Leftarrow$) If $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$, a subspace. If $W_2 \subseteq W_1$, then $W_1 \cup W_2 = W_1$, also a subspace.

Method 2: Apply the union-of-subgroups theorem

Theorem (Union of subgroups). If H, K are subgroups of a group G , then $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$.

($\Rightarrow$) View $(V, +)$ as an abelian group. Since $W_1, W_2 \leq (V, +)$ and $W_1 \cup W_2$ is assumed to be a *vector* subspace, it is in particular an additive subgroup of $(V, +)$. By the theorem,

$$W_1 \cup W_2 \text{ is a subgroup} \implies (W_1 \subseteq W_2 \text{ or } W_2 \subseteq W_1).$$

($\Leftarrow$) If $W_1 \subseteq W_2$ then $W_1 \cup W_2 = W_2$ is a subspace; if $W_2 \subseteq W_1$ then $W_1 \cup W_2 = W_1$ is a subspace.

Hence,

$$W_1 \cup W_2 \text{ is a subspace of } V \iff (W_1 \subseteq W_2 \text{ or } W_2 \subseteq W_1).$$

Question 6 (Subset of $\mathbb{R}^{2,2}$)

Problem

Determine whether

$$\left\{ \begin{pmatrix} a & b \\ 3b - 4c & a + 2 \end{pmatrix} \in \mathbb{R}^{2,2} \mid a, b, c \in \mathbb{R} \right\}$$

is a vector subspace of $\mathbb{R}^{2,2}$.

Solution

Let

$$S := \left\{ \begin{pmatrix} a & b \\ 3b - 4c & a + 2 \end{pmatrix} \mid a, b, c \in \mathbb{R} \right\}.$$

Method 1: Check whether the zero matrix is included

If S were a subspace of $\mathbb{R}^{2,2}$, it must contain the zero matrix

$$\mathbf{0} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Suppose $\mathbf{0} \in S$. Then there exist $a, b, c \in \mathbb{R}$ such that

$$\begin{pmatrix} a & b \\ 3b - 4c & a + 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Equating entries gives

$$a = 0, \quad b = 0, \quad 3b - 4c = 0, \quad a + 2 = 0.$$

But $a = 0$ and $a + 2 = 0$ cannot both hold, since $a + 2 = 0$ forces $a = -2$. This contradiction shows $\mathbf{0} \notin S$. Therefore S is not a subspace.

Hence the final answer is

S is not a vector subspace of $\mathbb{R}^{2,2}$.
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Question 7 (Direct sum complement in $\mathbb{R}^5$)

Problem

Define a vector subspace V of $\mathbb{R}^5$ as follows:

$$V := \{(a, b, a + b, a - b, 2a) \in \mathbb{R}^5 \mid a, b \in \mathbb{R}\}.$$

Find a vector subspace W of $\mathbb{R}^5$ such that $\mathbb{R}^5 = V \oplus W$.

Solution

Method 1: Basis extension via an explicit direct-sum decomposition

Write each $x \in V$ as

$$(a, b, a + b, a - b, 2a) = a(1, 0, 1, 1, 2) + b(0, 1, 1, -1, 0).$$

Hence

$$V = \text{span}\{v_1, v_2\}, \quad v_1 = (1, 0, 1, 1, 2), \quad v_2 = (0, 1, 1, -1, 0),$$

so $\dim V \leq 2$, and in fact v_1, v_2 are linearly independent (their first two coordinates form the standard basis of $\mathbb{R}^2$), hence $\dim V = 2$.

Define

$$W := \text{span}\{e_3, e_4, e_5\} = \{(0, 0, s, t, u) \in \mathbb{R}^5 \mid s, t, u \in \mathbb{R}\}.$$

Claim 1: $V \cap W = \{\mathbf{0}\}$. Let $x \in V \cap W$. Since $x \in W$, we have $x_1 = x_2 = 0$. Since $x \in V$, there exist $a, b \in \mathbb{R}$ such that

$$x = (a, b, a + b, a - b, 2a).$$

Comparing coordinates gives $a = x_1 = 0$ and $b = x_2 = 0$, hence $x = \mathbf{0}$. Therefore $V \cap W = \{\mathbf{0}\}$.

Claim 2: $V + W = \mathbb{R}^5$. Let $x = (x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5$. Set

$$v := (x_1, x_2, x_1 + x_2, x_1 - x_2, 2x_1) \in V,$$

and define $w := x - v$. Then

$$w = (0, 0, x_3 - (x_1 + x_2), x_4 - (x_1 - x_2), x_5 - 2x_1) \in W.$$

Thus $x = v + w$ with $v \in V$ and $w \in W$, proving $V + W = \mathbb{R}^5$.

Since $V + W = \mathbb{R}^5$ and $V \cap W = \{\mathbf{0}\}$, it follows that

$$\mathbb{R}^5 = V \oplus W.$$

Hence a valid choice is

$$W = \{(0, 0, s, t, u) \in \mathbb{R}^5 : s, t, u \in \mathbb{R}\} = \text{span}\{e_3, e_4, e_5\}.$$

Method 2: Kernel–image decomposition using a linear map (rank–nullity)

Define a linear map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$ by

$$T(x_1, x_2, x_3, x_4, x_5) := (x_1, x_2).$$

Then T is surjective, hence $\text{rank}(T) = 2$. Moreover,

$$\ker(T) = \{(0, 0, s, t, u) \in \mathbb{R}^5 : s, t, u \in \mathbb{R}\}.$$

We claim that $T|_V : V \rightarrow \mathbb{R}^2$ is an isomorphism. Indeed, for any $(a, b) \in \mathbb{R}^2$,

$$T(a, b, a + b, a - b, 2a) = (a, b),$$

so $T(V) = \mathbb{R}^2$ (surjectivity). Also, if $x \in V$ and $T(x) = (0, 0)$, then x has the form

$$x = (a, b, a + b, a - b, 2a) \in V \quad \text{with } (a, b) = (0, 0),$$

hence $x = \mathbf{0}$, so $T|_V$ is injective.

Now let $W := \ker(T)$. For any $x \in \mathbb{R}^5$, define $(a, b) := T(x) = (x_1, x_2)$, and let

$$v := (a, b, a + b, a - b, 2a) \in V.$$

Then $T(x - v) = T(x) - T(v) = (a, b) - (a, b) = (0, 0)$, hence $x - v \in \ker(T) = W$. Therefore $x = v + w$ for some $v \in V$, $w \in W$, so $\mathbb{R}^5 = V + W$.

Finally, if $x \in V \cap W$, then $x \in V$ and $x \in \ker(T)$ implies $T(x) = (0, 0)$. Since $T|_V$ is injective, this forces $x = \mathbf{0}$. Thus $V \cap W = \{\mathbf{0}\}$, and hence $\mathbb{R}^5 = V \oplus W$.

Consequently,

$$\boxed{W = \ker(T) = \text{span}\{e_3, e_4, e_5\}.$$